BGR modelling activities under the EURAD and CD-A Projects

Mont Terri Technical Discussion

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Overview of the recent works

1. BGR: Test of Constitutive Models (Shuang Chen)

- 2D model (Test of different solvers, computational efficiency analyse)
- Evaluation of strains
- Comparison with 3D model (Tuanny) and extensometer measurements
- Next: test of constitutive models

2. BGR–EDF (France) Collaboration

- Meeting on 5 March: exchange on the latest modelling results
- Coordination of the next work focus

Next focus: Comparison with Measurements:

- Displacements / Strains (Extensometer measurements)
- Pore pressure (Mini-Piezometer measurements)
- Pore pressure in desaturated zone (Suction measurments)

Approach:

- Stepwise benchmarking approach

Recent modelling works

- 2D model (Test of different solvers, computational efficiency analyse)
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Testing the 2D Model in OGS6
Linear Solver behavior

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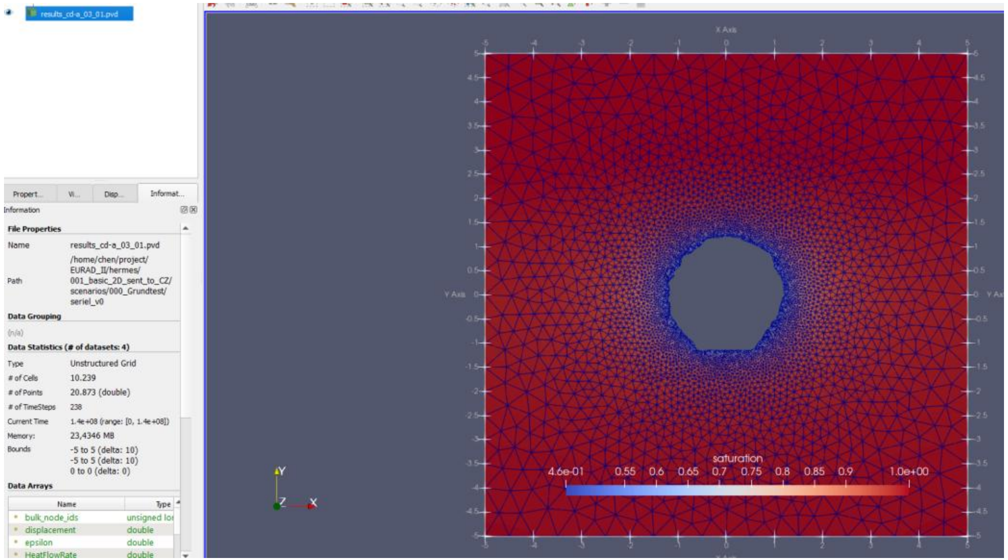
Scenario 1: 2D model with
precise surface geometry.

OGS6 process:
Thermal–Richards Flow–Mechanical (TRM)

Solver type:

Serial: PardisoLU, BiCGILUT

Parallel: MUMPS, HYPRE, GMRES, CPardiso.



Testing the 2D Model in OGS6
Linear Solver behavior

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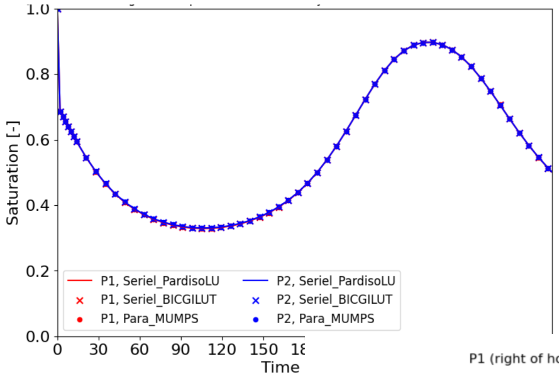
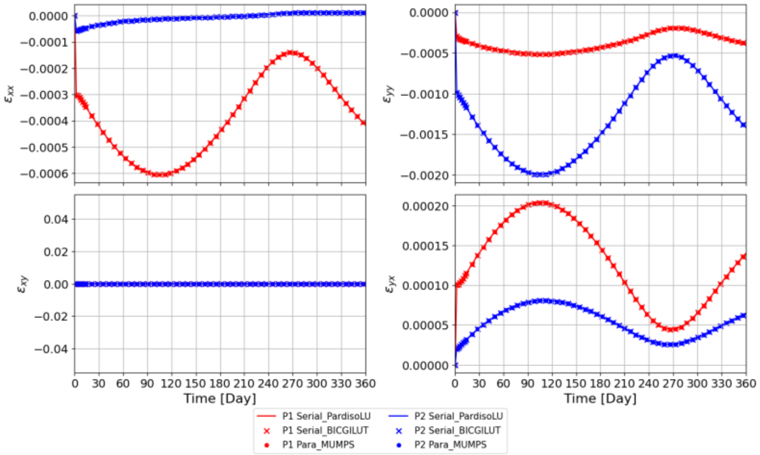
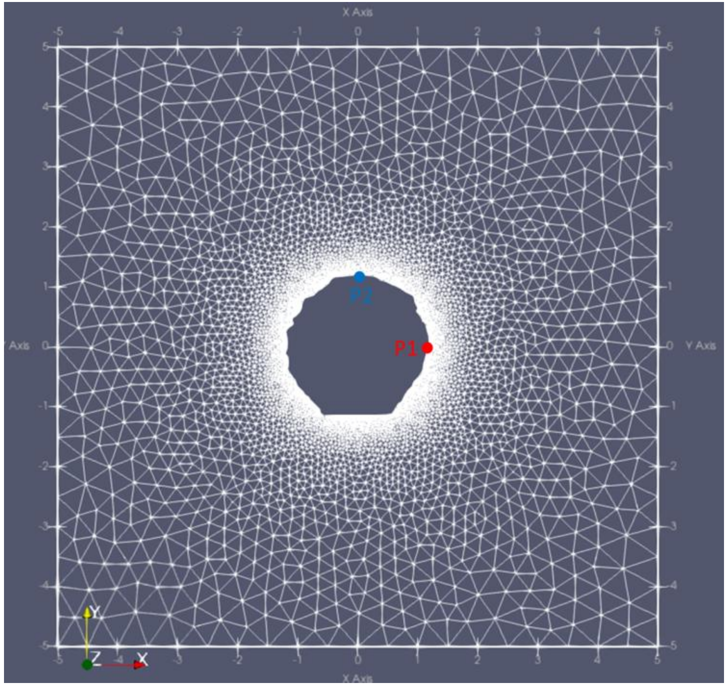
Verification of result consistency across solvers



Epsilon evolution at selected points
P1 (right of hole boundary – red), P2 (top of hole boundary – blue)

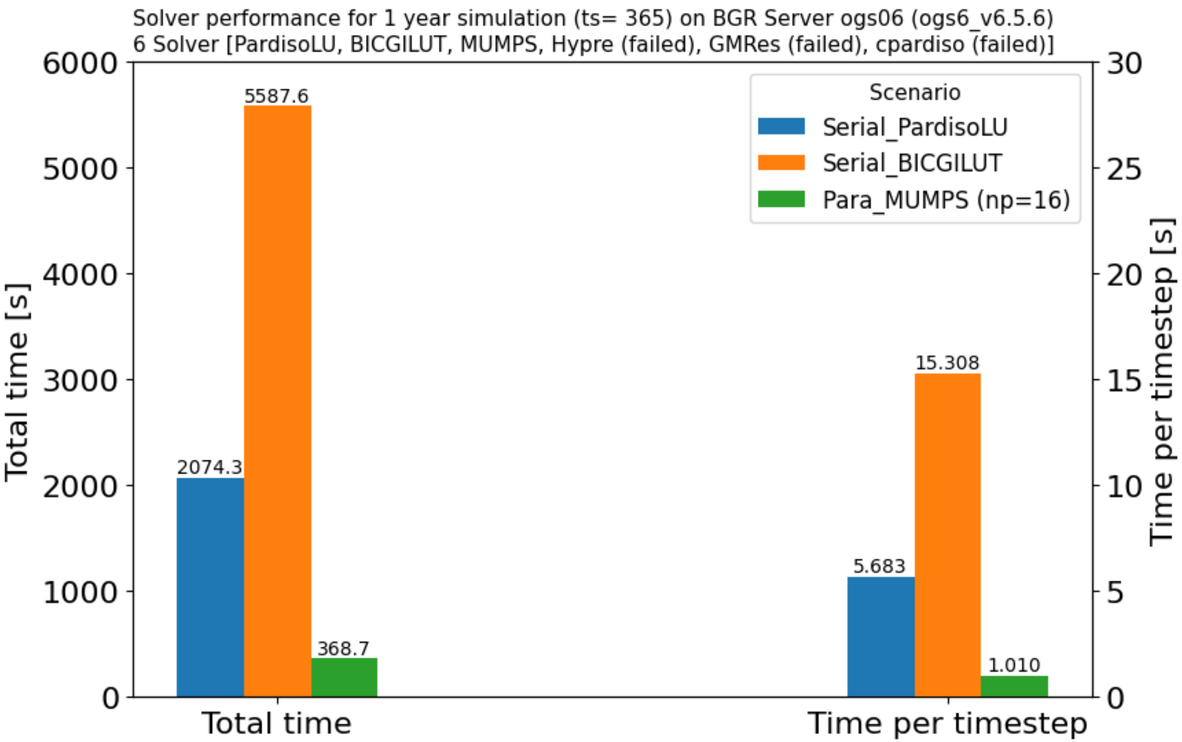


Output points for comparison



Testing the 2D Model in OGS6
Linear Solver behavior

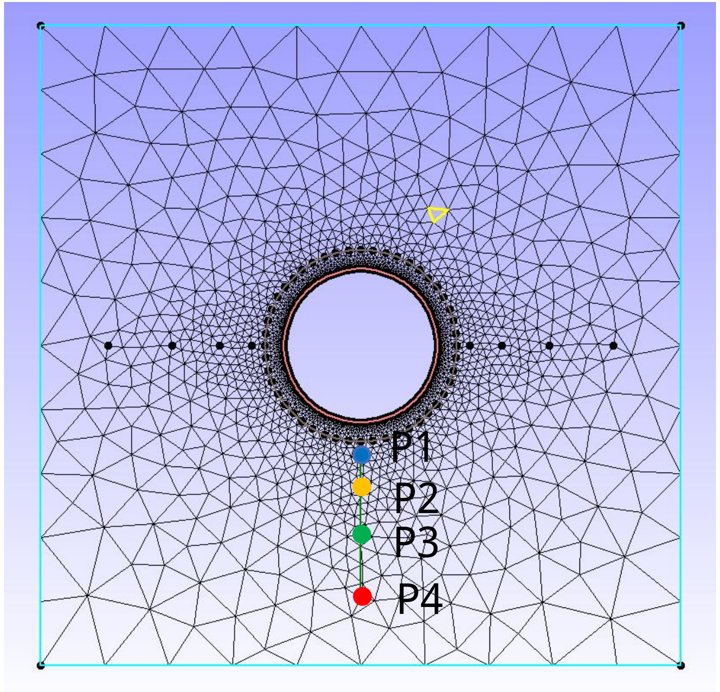
Comparison of computational cost across solvers



Testing the 2D Model in OGS6

Scenario 2: 2D model with simplified perfect ring geometry.

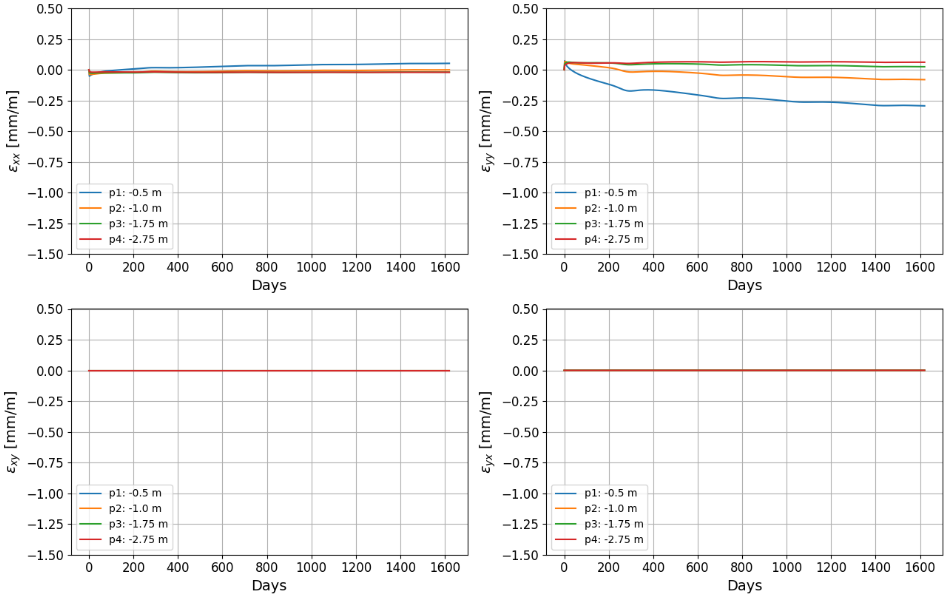
Mesh tool: GINA

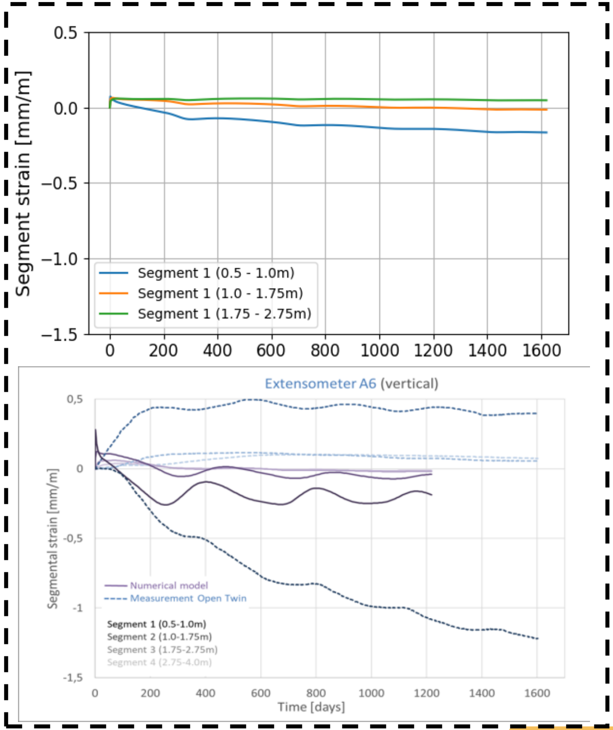


Model Setup:
Points: 3,620
Cells: 5,879
DOF: 14,480
Mesh resolution on the inner ring: 0.03 m.

Testing the 2D Model in OGS6

Preliminary comparison of results with measurements





BGR–EDF Collaboration

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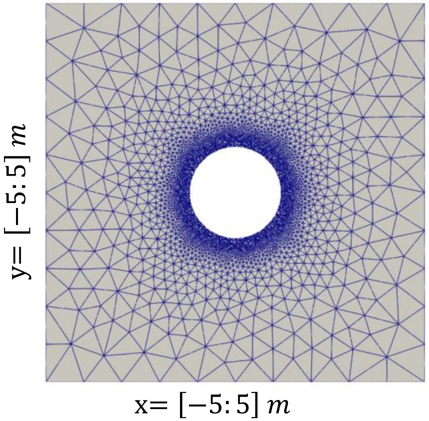
Stepwise benchmarking approach

Scenario Overview

Sce#1

2D Simplified Model

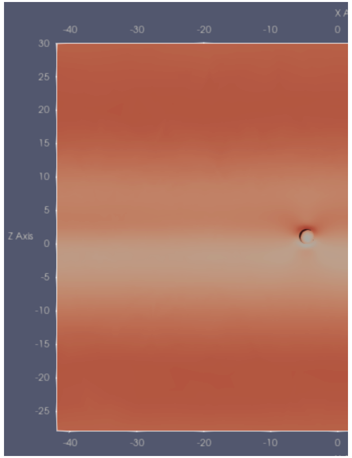
Geometry: 10m x 10m



Sce#2

2D Slice-Based Model

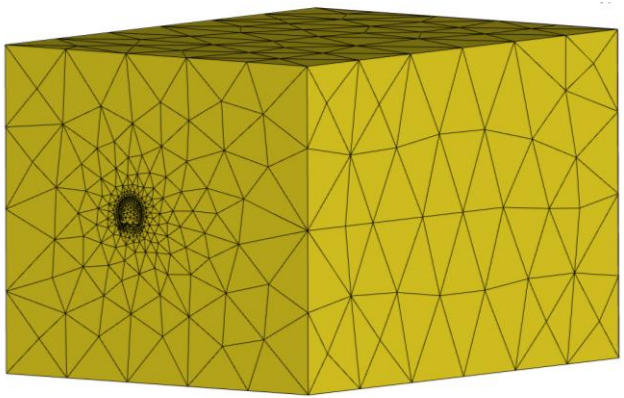
Geometry: slice of the 3D model



Sce#3

3D Model

Open and closed niches and gallery



Detail Information: Excel File

2D_Model_Setup_v1.xlsx

Motivation

A stepwise benchmarking approach from a simple 2D model to the 3D in-situ model provides a practical basis for benchmarking, later constitutive model testing, and finally comparison with extensometer and pore pressure measurement data.

The intermediate 2D slice-based model serves as a practical compromise between the simplified 2D case and the full 3D model, retaining the practical advantages of 2D model assessment while allowing testing under conditions that more closely approximate the 3D model. In particular, it enables consideration of the in-situ stress state, which will likely require explicit simulation of the excavation process.

Sce#1: 2D Simplified Model

Process: THERMO_RICHARDS_MECHANICS

Initial conditions (IC)

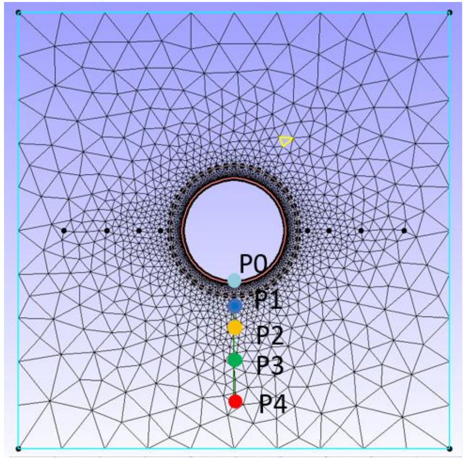
- Homogeneous

Hydraulic boundary conditions (BC)

- At the wall of the niche

$RH \rightarrow p_L$ (Open_niche_seasonal_sinus_curve)

Mechanical boundary conditions (BC)



Material properties

Category	Parameter	Value
Porosity	phi	0.105
Intrinsic Permeability	k	1.00E-19 0.00 0.00 0.40E-19 m2
Biot	alpha	1
Elastic	E	13.8 / 6.0 / 13.8 GPa
Elastic	G	4.79 / 2.46 / 4.79 GPa
Elastic	nu	0.44 / 0.22 / 0.44
Solid	density	2520 kg/m3
Fluid	density	linear (T,p dependent)
	reference_value	1095 kg/m3
Fluid	viscosity	1e-3 Pa·s
Retention	model	Van Genuchten
	Residual liquid saturation (S_lr)	0.1
	Residual gas saturation (S_gr)	0.0
	Exponent	0.45
	Air-entry pressure (p_b)	10e6 Pa
Swelling	model	SaturationDependentSwelling
	reference_value	1.0E+06 Pa
Solid	thermal conductivity	2.156 W/mK
Solid	heat capacity	1254.74 J/kgK
Solid	thermal expansion	1254.74 J/kgK
Fluid	heat capacity	4160 J/kgK
Fluid	thermal conductivity	0.6 W/mK

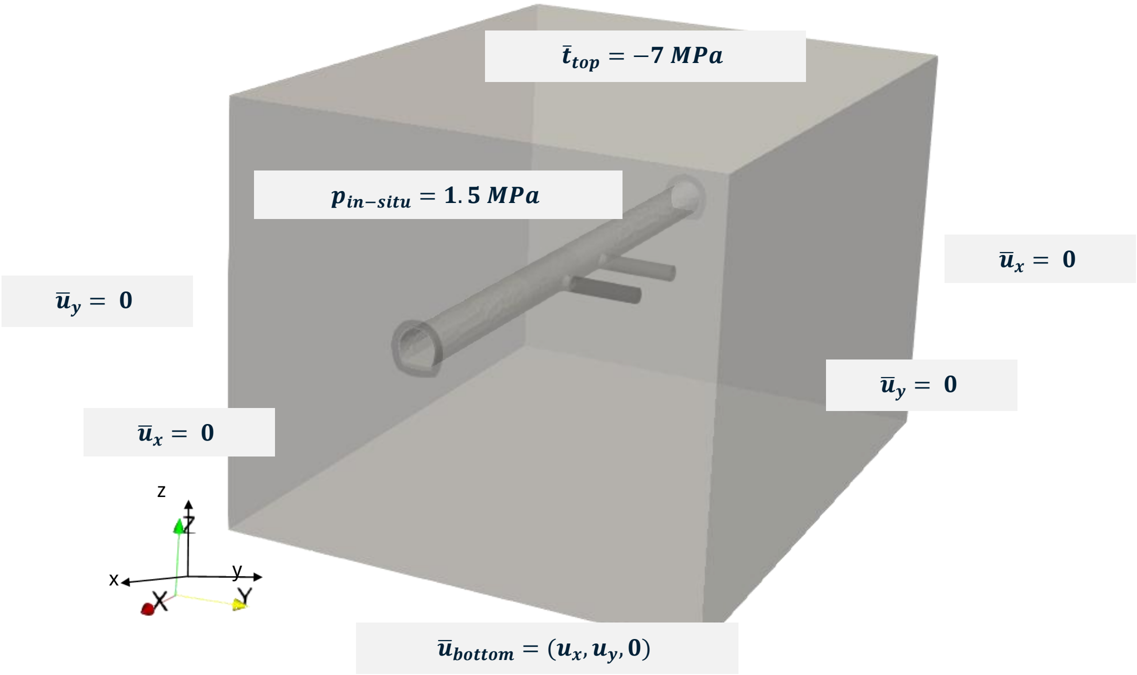
Expected outputs:

Point ID	Location	Variables
P0	0.0, -1.15	liquid pressure, displacement, saturation (pL, u, S)
P1	0.0, -1.65	pL, u, S
P2	0.0, -2.15	pL, u, S
P3	0.0, -2.9	pL, u, S
P4	0.0, -3.9	pL, u, S

Sce#2: 2D Slice-Based Model

Detailed model setup will follow after the proposed scenarios have been discussed and agreed upon.

Sce#3: 3D Model



Summary and next todos

2D Model
Model precise surface geometry: Tested solver behavior and efficiency.
Model with perfect ring geometry: Simulated strain showed discrepancies vs. measurements. Model setup is prepared for the next benchmarking scenario.

Modelling Strategy
Stepwise benchmarking: Simple 2D → Complex 3D in-situ model. Finally targets to compare with extensometer and pore pressure measurement data.

Next steps
- BGR internal: test of constitutive models.
- Collaboration**:** Benchmark BGR vs. EDF modelling.